Experiment 2 : Project Design

Problem Statement:

People with speech and hearing disabilities often struggle to communicate in real time during conversations, online classes, interviews, and video calls because most systems are designed only for spoken language. Existing sign-language recognition solutions are either expensive, require special hardware such as sensor gloves, depend on cloud servers, or are too complex to build and deploy quickly. There is a need for a simple, low-cost, real-time system that can convert Indian Sign Language (ISL) gestures into spoken words using only a webcam and a laptop.

Solution:

AccessTalk is a local Python-based application that captures hand gestures through a webcam, detects 21 hand landmarks using MediaPipe, converts those landmarks into 42 coordinate values, and uses a trained Random Forest model to recognize one of 15 ISL words. Once a gesture is recognized with sufficient confidence, the system displays the predicted word on the screen and speaks it aloud using a text-to-speech engine. The complete system works offline, requires no server or internet connection, and runs in real time on a normal laptop.

Functional Requirements

- The system shall capture live video from the user's webcam.
- The system shall detect a single hand and extract 21 hand landmarks from each frame.
- The system shall classify the extracted landmarks into one of the 15 supported ISL words.
- The system shall display the predicted word and confidence score on the screen.
- The system shall convert the recognized word into speech using an offline text-to-speech engine.

Non-Functional Requirements

- The system shall perform recognition in real time with an end-to-end latency of less than 50 ms.
- The system shall work completely offline without requiring internet access or cloud services.
- The system shall run on standard laptop hardware without needing a GPU or external devices.
- The system shall be easy to install and execute using a single Python-based application.
- The system shall provide reliable recognition accuracy of approximately 85–95% for the supported gestures under proper lighting conditions

Work items - Boards

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53 As a system, a trained model classifies ISL gestures with 90%+ accuracy

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Updated by Suchit Prabu V: Apr 7

State: New | Area: Access Talk Project

Reason: New | Iteration: Access Talk Project\Iteration 1

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Description

Suchit trains a Random Forest classifier on the collected CSV, evaluates accuracy, and exports model.pkl for the backend.

Acceptance Criteria

- Accuracy is 90%+ on test set
- model.pkl is exported and uploaded to Google Drive
- Classification report printed for all 15 gestures

Discussion

Planning

Story Points: 12

Priority

Risk: 1 - High

Classification

Value area: Business

Deployment

To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting

Development

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Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.

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